

NLU Project – Part 1: LM

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1. Introduction

This report outlines my work and findings for Part 1 of the NLU course project at the University of Trento, which focused on language modeling for next-word prediction. The project is divided into two main sections: Part A and Part B.

- **Part A** enhances the baseline Recurrent Neural Network (RNN) language model by replacing the RNN with a Long Short-Term Memory (LSTM) network, adding dropout layers, and switching from Stochastic Gradient Descent (SGD) to the AdamW optimizer.
- **Part B** builds on the best-performing vanilla LSTM model from Part A, investigating three advanced regularization methods: Weight Tying, Variational Dropout, and Non-Monotonically Triggered Averaged SGD (NT-AvSGD).

In both parts, key hyperparameters—including learning rate, hidden size, embedding dimension, and batch size—were systematically tuned to minimize perplexity (PPL).

2. Methods

All modifications were applied incrementally, with performance monitored at each stage using loss curves. Early stopping with a patience of 3 epochs was used throughout, except when employing NT-AvSGD. Additionally, a test mode was implemented to re-evaluate saved models and verify the logged results.

In Part A, improvements to the baseline RNN included:

1. Replacing it with an LSTM.
2. Adding two dropout layers—one after the embedding layer and another before the final output layer.
3. Switching the optimizer from SGD to AdamW.

For Part B, starting from the best-performing vanilla LSTM in Part A, the following regularization techniques proposed by Merity et al were implemented:

- **Weight Tying:** Sharing weights between the embedding and output layers to reduce parameters and enforce consistency. When layer dimensions differed, a projection layer was introduced to accommodate mismatched sizes.
- **Variational Dropout:** Applying the same dropout mask at every time step to maintain temporal consistency and improve generalization.
- **NT-AvSGD:** An advanced variant of SGD where averaging of model weights begins only after validation perplexity fails to improve over a predefined number of evaluations. As detailed in the reference paper, the switch is triggered when validation loss surpasses the minimum value within a non-monotonic loss window. After activation, weights are averaged until training concludes, stabilizing training and

boosting generalization. This method was adapted from the paper's pseudocode into a Python implementation.

3. Results

Extensive hyperparameter tuning was conducted to enhance model performance, with perplexity (PPL) serving as the primary evaluation metric. The goal was to reduce PPL as much as possible, improving next-word prediction accuracy. The final test perplexities for the best models in each configuration are summarized below.

Part A: Starting from the baseline LSTM model, incremental improvements yielded consistent performance gains:

- The plain LSTM model achieved a test perplexity of **160.00**.
- Introducing dropout significantly improved generalization, lowering PPL to **102.43**.
- Replacing SGD with the AdamW optimizer provided a further modest improvement, reaching a PPL of **101.37**.

These results highlight the importance of regularization and optimizer choice for stabilizing training and reducing overfitting.

Part B: Building upon the best vanilla LSTM from Part A, three advanced regularization and optimization techniques were evaluated:

1. **Weight Tying:** Sharing weights between the embedding and output layers reduced parameters and improved performance, resulting in a test perplexity of **117.98**.
2. **Variational Dropout:** Applying a consistent dropout mask across time steps further reduced overfitting and achieved the best test perplexity of **95.79**.
3. **Non-Monotonically Triggered Averaged SGD (NT-AvSGD):** This optimizer variant stabilized training by averaging weights only after validation loss plateaus. However, it yielded a slightly higher test perplexity of **96.02**, possibly due to sensitivity to hyperparameter settings or the non-monotonic interval choice.

Overall, the combination of structural regularization (Weight Tying, Variational Dropout) and advanced optimization (NT-AvSGD) demonstrated clear benefits, with Variational Dropout achieving the lowest perplexity on the test set.

Analysis of NT-AvSGD Performance:

Although NT-AvSGD is designed to stabilize training and improve generalization by averaging model weights only after validation loss plateaus, its slightly higher perplexity (**96.02**) compared to Variational Dropout (**95.79**) suggests some sensitivity to hyperparameter settings. Specifically, the choice of the non-monotonic interval and logging frequency can impact when averaging is triggered. If averaging starts too late or too early, the benefits may be diminished, leading to suboptimal convergence. Additionally, the relatively small difference in perplexity indicates that NT-AvSGD remains competitive and may benefit from further fine-tuning or integration with other regularization techniques.

Table 1: Part A – Incremental Model Improvements with Hyperparameters and Test Perplexities

Model Configuration	Learning Rate	Batch Size (Test)	Hidden/Embedding Size	Dropout	Test Perplexity
LSTM (baseline)	1.0	128	256 / 256	–	160.00
+ Dropout	1.0	128	256 / 256	0.4	102.43
+ AdamW optimizer	0.001	128	256 / 256	0.4	101.37

Table 2: Part 2B – Regularization Techniques with Hyperparameters and Test Perplexities

Model Configuration	Learning Rate	Batch Size (Test)	Hidden/Embedding Size	Dropout	Test Perplexity
Vanilla LSTM	1.0	128	256 / 256	–	160.00
+ Weight Tying	1.0	128	256 / 256	–	117.98
+ Variational Dropout	1.0	128	256 / 256	0.4	95.79
+ NT-AvSGD	1.0	128	256 / 256	0.4	96.02